DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISHAL

FINAL EXAMINATION-2021

Course Title: Theory of Computation
Course Code: CSE-3203
3rd Year, 2nd Semester, Session: 2018-19

Time: 3 hours **Marks:** 60

Answer any Five (5) Questions from the followings.

1. a) What is Automata Theory? Describe the applications of Finite Automata. 3
   b) Explain the work of lexical analyzer with an on/off switch. 2
   c) The Chomsky Hierarchy is a containment hierarchy of classes of formal languages. Describe with suitable figure(s). 4
   d) Analyze the terms: Theorems, Lemmas and Corollaries. 3

2. a) Describe the basic functionalities of pumping lemma. 3
   b) Use the pumping lemma to prove that the language $A = \{ 0^{2n} \mid ^{3n}0^{n} \mid n \geq 0 \}$ is not context free. 6
   c) Define push down automata with example. 3

3. a) Describe Deductive proof. Prove that, if x is the sum of the squares of four positive integers, then $2^x \geq x^2$. 4
   b) Briefly explain the configuration of a TM. 3
   c) Let **x** be a real number. Then prove that, $\lfloor \mathbf{x} \rfloor = \lceil \mathbf{x} \rceil$ if and only if **x** is an integer. 2
   d) For all $\mathbf{n} \geq \mathbf{0}$, prove that $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$. 3

4. a) Describe with examples: advantages of DFA. 4
   b) Construct a DFA for the following language: 5
   Let, $\Sigma = \{0, 1\}$, L = {w | w is a binary string that has even number of 1s and even number of 0s}.
   c) Construct a NFA for the following: Strings where the first symbol is present somewhere later on at least once. 3

5. a) Build an NFA for the following language: L = { w | w ends in 101} 4
   b) Describe the Advantages & Caveats for NFA. 3
   c) Convert the NFA from Question 5(a), to DFA. 5

6. a) Describe the necessities of explicit ε-transitions in finite automata. 2
   b) Build an ε-NFA for the following language: 4
   L = {w | w is empty, or if non-empty will end in 11}
   c) Convert ε-NFA to DFA based on Question 6(b) 6

7. a) What is regular expression? Describe the operations of regular expressions. 3
   b) Prove that- if L=L(A) for some DFA A, then there is a regular expression R such that L=L(R). 3
   c) Convert the following DFA to an equivalent regular Expression. 3



   d) Define- i) Associativity ii) Identity and iii) Distributive Law 3

8. a) Regarding conversion of DFA from NFA, a bad case for the Subset Construction is occurred. Analyze this using the Pigeonhole Principle. 6
   b) Briefly describe, how we can eliminate ε- transitions? 4
   c) What are the uses of ε-Transitions? 2

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